NICOLÁS CARO

TRANSPORT ENGINEER



CONTACT

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Linkedin

Portfolio

SKILLS

- Transport & Mobility Analysis
- Origin–Destination (OD) Matrix Development
- Spatial Analysis & GIS (ArcGIS, QGIS, GeoPandas)
- Data Analytics & Visualization (Python, Power BI, R Studio, SQL)
- Big Data Processing (Pandas, NumPy)
- Urban & Transport Planning Support
- Quantitative Modeling
- Scenario Analysis & Decision Support
- Technical Reporting & Documentation
- Policy-Oriented Data Interpretation
- Excel & VBA Automation
- Transport Modeling Tools (EMME, PTV Vissim)
- BIM & Infrastructure Tools (Revit, Navisworks)

LANGUAGES

- English (Fluent)
- Spanish (Fluent)
- French (B2)

TRAINING COURSES

- Data Spaces for Mobility
- PTV Vissim Modelling
- Modal Split Model Construction

PROFILE

I'm a Civil Engineer with a Master's in Data Science for Smart Mobility, passionate about using data to understand and improve how cities move. My experience includes developing OD matrix algorithms, mobility dashboards, and analytical tools for transport planning projects across Europe, Latin America, and the Middle East.
With strong skills in Python, R, Power BI, and GIS, I enjoy connecting data, technology, and mobility to support smarter and more sustainable cities.

WORK EXPERIENCE

SENER 2022 - PRESENT

Transport Economics and Planning Engineer — Barcelona, Spain

- Developed a Python-based algorithm to infer public transport Origin–Destination (OD) matrices using the trip chaining method, integrating GIS-based zonification and interactive visualization dashboards in Power BI in the Basque Country.
- Led analytical tasks for large-scale mobility studies, including the design of geographical zonings and the generation of OD matrices using public transport validations and mobile phone data in Panama.
- Contributed to the development of Girona´s Sustainable Urban Mobility Plans (SUMP) by producing GIS-based maps, spatial analyses, and scenario evaluations for public and private transport systems.
- Supported feasibility and integration studies for innovative and disruptive mobility services, providing data-driven insights to inform strategic planning and decision-making in Saudi Arabia.
- Assisted as Lead Technical Advisor in the extension of the Red Line light rail in Jerusalem.

WSP 2021 - 2022

Transport Engineer — Bogotá, Colombia

- Led the realization of the multidisciplinary monthly report and KPI dashboards to review and analyze the performance and status of the project using R Studio and Power BI for the metro line 1 in Bogotá.
- Facilitated the compilation of all the databases of the project in order to have a complete picture of the items sent, received, or in progress.

EDUCATION

Master's Degree in Urban Mobility 2022 - 2024

Universitat Politècnica de Catalunya | Barcelona, Spain

Bachelor of Civil Engineering, Business Minor 2016 - 2021

Universidad de los Andes | Bogotá, Colombia

Intensive French Course 2015-2016

Université Grenoble | Grenoble, France